

Industrial Internship Report on

Python URL Shortener

Prepared by

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was a Python-based URL Shortener developed using Flask and SQLite. The application validates long URLs, generates unique short codes, stores URL mappings, redirects users to original destinations, tracks clicks, and provides URL history and management features.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

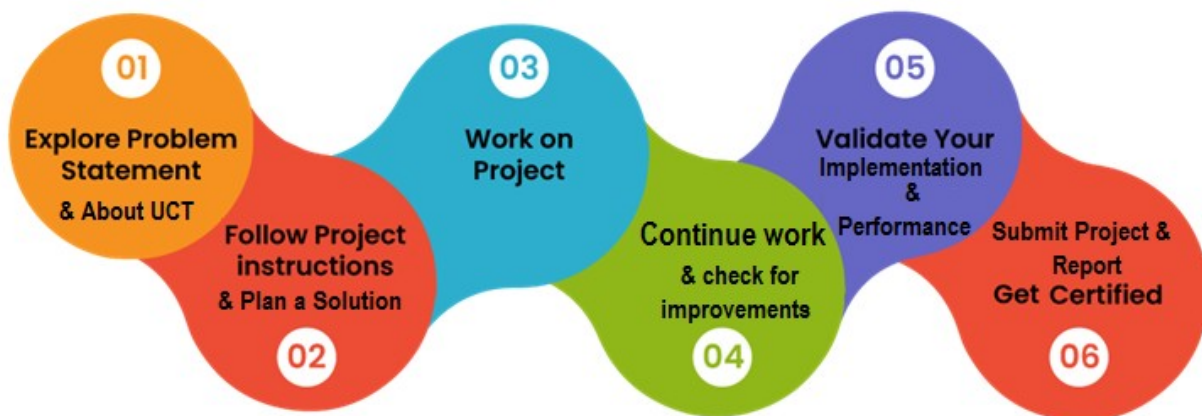
This six-week industrial internship focused on strengthening Python programming and practical software development skills. During the internship, I studied Python fundamentals, conditional statements, NumPy, Pandas, and related programming concepts. I also developed and improved a Python-based URL Shortener using Flask and SQLite, tested its functionality, and documented the project.

The internship provided practical exposure to software development beyond classroom learning. It helped me understand how programming concepts are applied to build, test, debug, document, and present a working application, supporting my career development in software engineering.

The project addresses the difficulty of sharing long and inconvenient web URLs. The proposed Python URL Shortener accepts a valid long URL, generates a unique short code, stores the URL mapping in SQLite, and redirects users from the shortened URL to the original destination.

Upskill Campus (USC), The IoT Academy, and UniConverge Technologies Pvt Ltd (UCT) provided the internship structure, learning resources, project work, and practical exposure described in the internship program.

The program was planned through weekly learning resources, assessments, practical programming activities, project development, testing, and documentation. The weekly activities progressed from Python fundamentals to data-related Python topics and practical project work.



The internship improved my Python programming, web development, database handling, debugging, testing, and documentation skills. Developing the URL Shortener helped me connect Python concepts with a real working application.

I sincerely thank Upskill Campus, The IoT Academy, UniConverge Technologies Pvt Ltd, my faculty members, mentors, and everyone who supported and guided me directly or indirectly throughout the internship.

My message to juniors and peers is to use internships as opportunities to practice consistently, build real projects, test their work carefully, and document what they learn. Practical implementation makes technical concepts easier to understand and improves confidence.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and Rol.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end etc.**



i. UCT IoT Platform

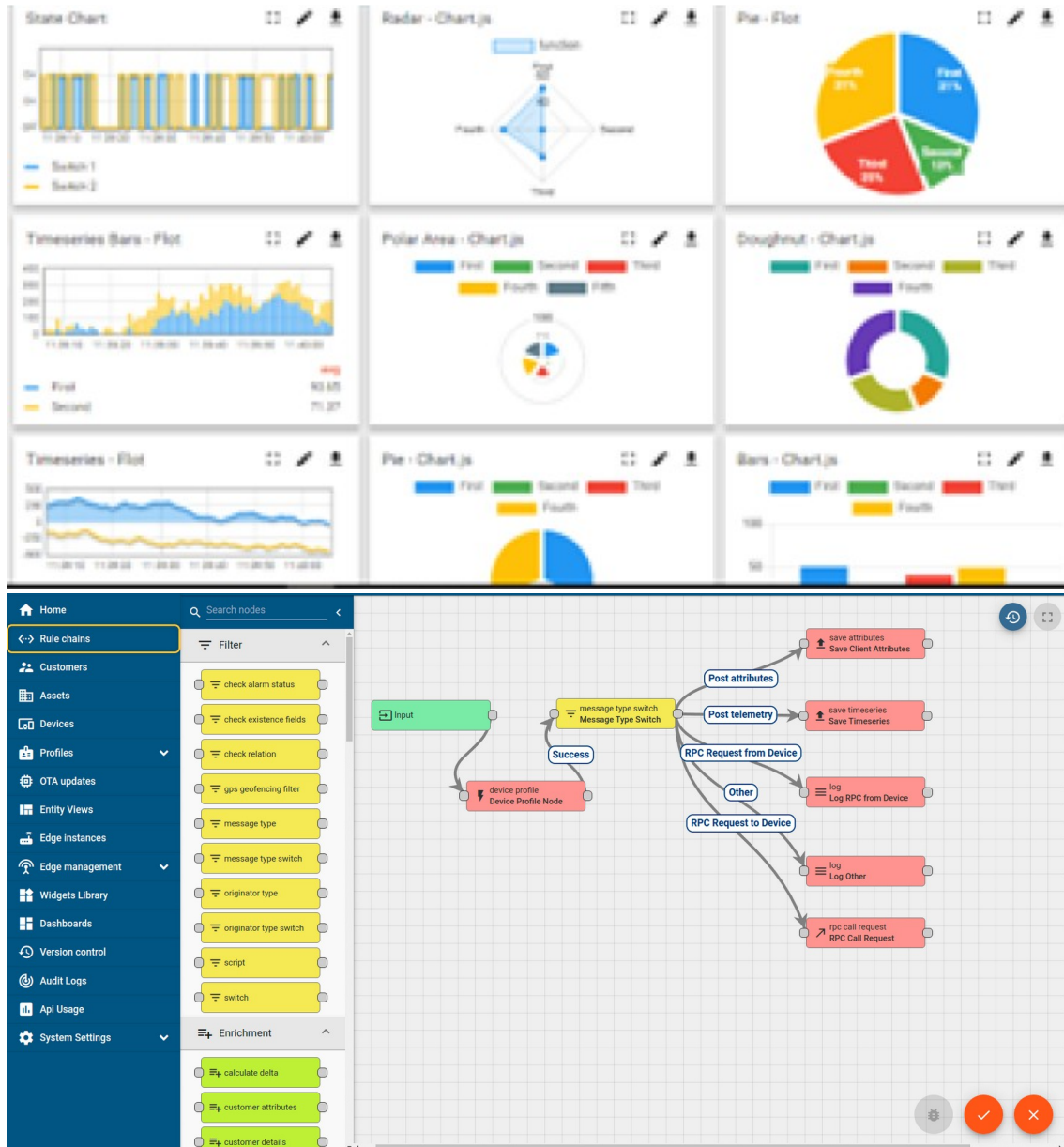
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting

- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





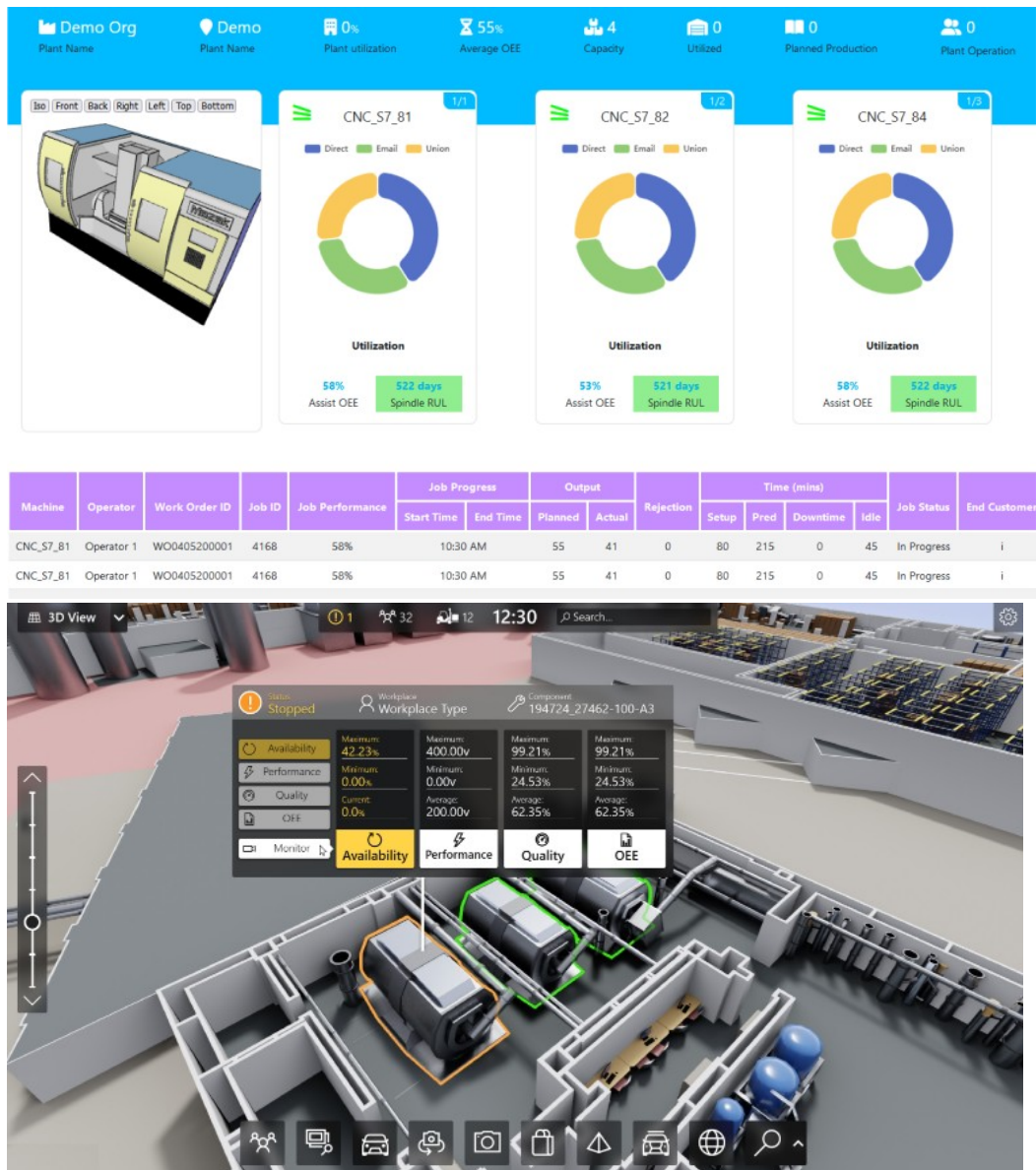
ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

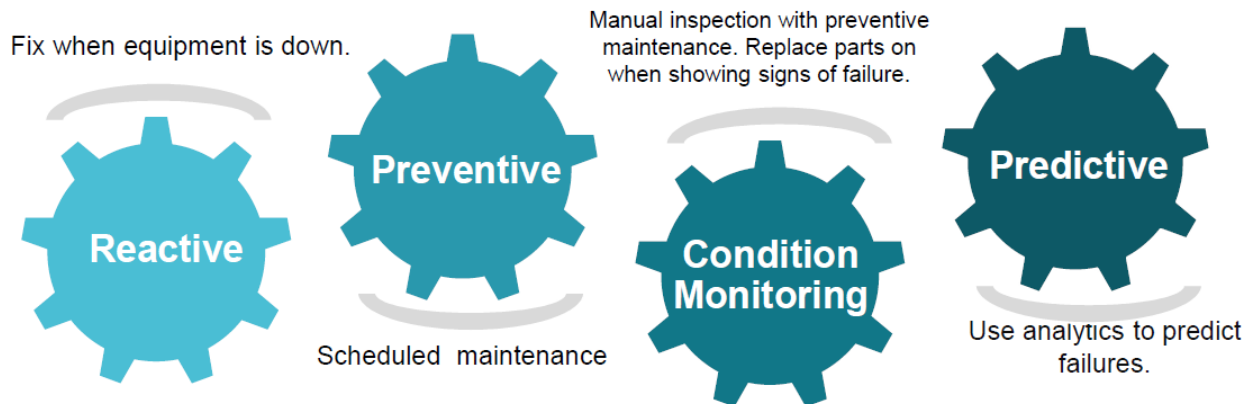


iii. LoRaWAN-based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

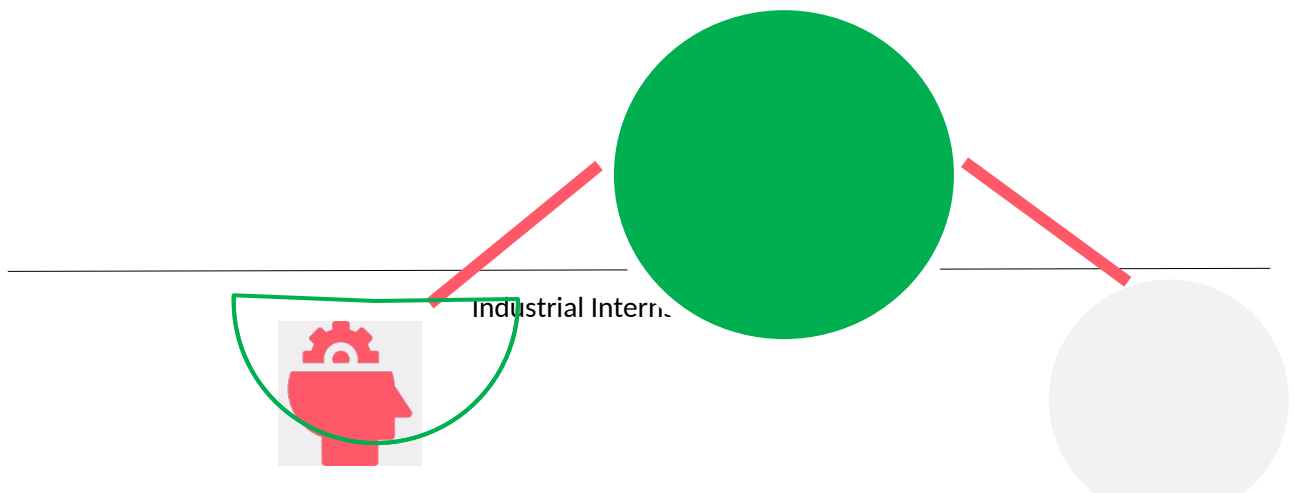
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.

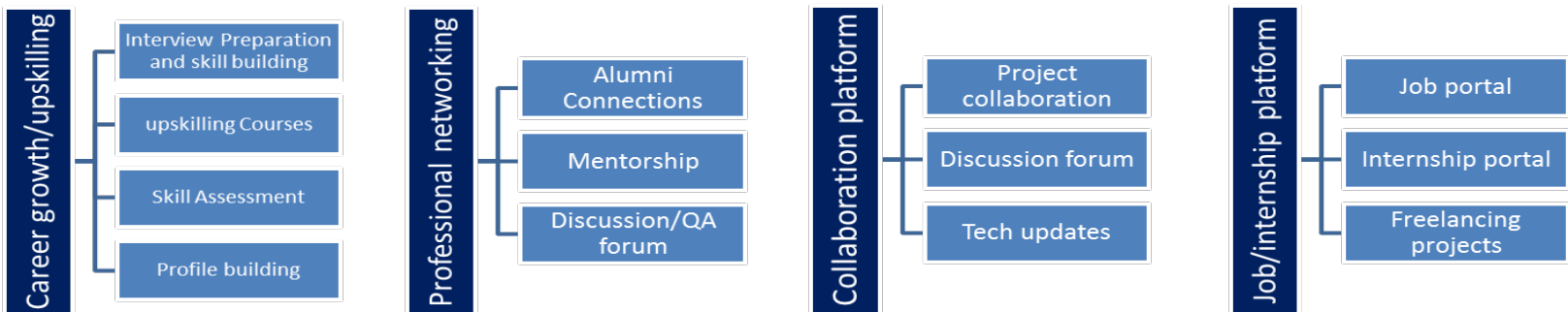


2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.





2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Upskill Campus Internship learning resources and weekly course materials.
- [2] Python documentation and Flask documentation used during project development.
- [3] GitHub repository containing the project source code and report.

2.6 Glossary

Terms	Acronym
API	Application Programming Interface
CSS	Cascading Style Sheets
Flask	Python web framework
HTTP	Hypertext Transfer Protocol
SQLite	Lightweight relational database
URL	Uniform Resource Locator

3 Problem Statement

In the assigned problem statement, the project focuses on developing a web-based URL Shortener that converts long web addresses into shorter, manageable links while preserving the ability to redirect users to the original URLs.

The system should accept a valid URL, validate the input, generate a unique short code, store the original URL and short code mapping, return the shortened URL, redirect users to the original URL when the short link is accessed, and maintain click statistics. The application also provides URL history, search, copy, and delete functionality.

4 Existing and Proposed solution

Existing URL-shortening services provide short links and redirection, but external services may provide limited control over stored mappings and may not demonstrate the internal implementation to a learner. For this internship project, a local Flask and SQLite implementation was developed to provide direct control over URL storage, validation, redirection, and usage tracking.

The proposed solution is a Python Flask web application with SQLite storage. It validates URLs, checks for duplicate entries, generates unique short codes, stores URL mappings, redirects users, tracks clicks, and provides a history dashboard.

The project adds practical features such as duplicate URL handling, click tracking, URL history, search/filtering, copy-to-clipboard, deletion, and dashboard statistics in a lightweight application.

4.1 Code submission (GitHub link):
<https://github.com/jsanthosh20/upskillcampus/blob/main/app.py>

4.2 Report submission (GitHub link):
https://github.com/jsanthosh20/upskillcampus/blob/main/URLShortener_SANTHOSH_USC_UCT.pdf

4.3

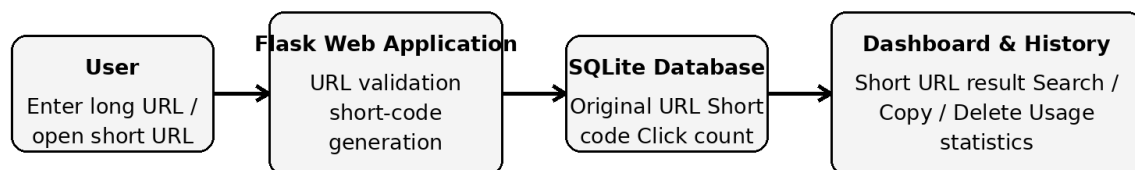
5 Proposed Design/ Model

The system starts when a user enters a long URL through the web interface. Flask receives the request and validates the URL. The application checks SQLite for an existing mapping; if none exists, it generates a unique short code and stores the mapping. The short URL is displayed to the user. When the short URL is accessed, Flask retrieves the original URL, increments the click count, and redirects the user to the original destination.

5.1 High Level Diagram

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

High Level Architecture - Python URL Shortener



Request flow: User → Flask → SQLite → Result / Redirect

Short URL access: User → Flask → SQLite lookup → Original URL

5.2 Low Level Design

Backend routes: / (dashboard), /shorten (POST URL creation), /<short_code> (redirect), /history (URL history), and /delete/<short_code> (delete). The SQLite table stores id, original_url, short_code, and clicks. Parameterized SQL queries are used for database operations.

5.3 Interfaces and Data Flow

The application uses HTTP requests between the browser and Flask routes. Form input is validated before database operations. SQLite stores the original URL, short code, and click count. The main flow is: Input URL → Validate → Check Duplicate → Generate Short Code → Store Mapping → Display Short URL → Redirect → Update Click Count.

6 Performance Test

Performance testing was used to verify that the application correctly handles the main URL-shortening workflow and common user inputs. The tests focused on correctness, validation, duplicate handling, redirection, database storage, click tracking, and dashboard functions.

The main constraints considered were valid URL input, uniqueness of short codes, persistent storage, correct redirection, and accurate click counting.

URL validation is performed before storage. The application checks existing URLs to handle duplicates and checks generated short codes for uniqueness. SQLite provides persistent storage, while the redirect route updates click counts after a successful lookup.

The implemented test cases covered valid URLs, invalid inputs, duplicate URLs, redirection, click counting, history, search, copy, and deletion. The expected behavior was verified during local testing of the application.

For this web application, the relevant constraints are input validity, database consistency, short-code uniqueness, response correctness, and reliable redirection rather than hardware-specific constraints such as power consumption.

Large-scale deployment would require additional load testing, stronger database indexing, rate limiting, authentication, monitoring, and production-grade hosting. These improvements can help maintain reliability as the number of URLs and users increases.

6.1 Test Plan / Test Cases

6.2 Test Procedure

The application performed as expected for the tested local use cases. Valid URLs were shortened, invalid inputs were rejected, duplicate URLs returned the existing short link, shortened URLs redirected correctly, and click counts were updated. History, search, copy, and delete functions were also verified.

6.3 Performance Outcome

Test Case	Input / Action	Expected Result	Outcome
1	Valid URL	Short URL generated and stored	Pass
2	Invalid URL	Validation message displayed	Pass
3	Duplicate URL	Existing short URL returned	Pass
4	Open short URL	Redirects to original URL	Pass
5	Open short URL multiple	Click count increases	Pass

	times		
6	History / Search	Stored URLs can be viewed and filtered	Pass
7	Copy / Delete	URL can be copied or removed	Pass

7 My learnings

The internship strengthened my understanding of Python programming, Flask web development, SQLite database operations, HTML/CSS, JavaScript, debugging, testing, and project documentation. These skills are directly useful for entry-level software engineering and full-stack development roles. The project also improved my ability to translate requirements into a working application.

8 Future work scope

Future enhancements can include user authentication, custom short URLs, QR-code generation, expiry dates, rate limiting, analytics dashboards, API endpoints, stronger security controls, automated testing, and cloud deployment. These additions can make the system more suitable for larger-scale use.